

Multi-Node Docker Swarm Cluster with Distributed Service Orchestration

```
Terminal
PS C:\Users\sreev> docker swarm init
Swarm initialized: current node (vpo87xg232tywt2gg322bt9e4) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-5bgn26n2382oon5thp3dw6kljnx1nyhp2xaaupkyznftckj8-bs9z8hsl312l5scc7dnnoxrto 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\sreev> docker service create --name web --replicas 3 -p 8882:80 nginx
b8zwj9xroqatlow31k9wkon9w
overall progress: 3 out of 3 tasks
1/3: running [=====>]
```

```
Terminal
b8zwj9xroqatlow31k9wkon9w
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service b8zwj9xroqatlow31k9wkon9w converged
PS C:\Users\sreev> docker service ls


| ID           | NAME | MODE       | REPLICAS | IMAGE        | PORTS          |
|--------------|------|------------|----------|--------------|----------------|
| b8zwj9xroqat | web  | replicated | 3/3      | nginx:latest | *:8882->80/tcp |


PS C:\Users\sreev> docker service ps web


| ID           | NAME  | IMAGE        | NODE           | DESIRED STATE | CURRENT STATE         | ERROR | PORTS |
|--------------|-------|--------------|----------------|---------------|-----------------------|-------|-------|
| 1uohap99j3u5 | web.1 | nginx:latest | docker-desktop | Running       | Running 5 minutes ago |       |       |


RAM 4.45 GB CPU 16.47% Disk: 69.10 GB used (limit 1006.85 GB)
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```
Terminal
PS C:\Users\sreev> docker run -d --name worker1 --privileged --network net1 docker:dind
docker: Error response from daemon: Conflict. The container name "/worker1" is already in use by container "259357f987dc40186cf9096320cf5b2aea5ba87657f94ae9fa1a8ab79d4e1158". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\Users\sreev> docker run -d --name workerb --privileged --network net1 docker:dind
6ff618d17e5a3eacd8774a145af350910fba7efcf25ed08898a999fae68dc4036
PS C:\Users\sreev> docker run -d --name workerc --privileged --network net1 docker:dind
c5ca2cc864d6d286f93a5be980c67283feeb760a83702c1a88702c6b3dce242
```

```
Terminal
PS C:\Users\sreev> docker exec -it workerb sh
/# ^C

/# docker swarm join --token SWMTKN-1-5bgn26n2382oon5thp3dw6kljnx1nyhp2xaaupkyznftckj8-bs9z8hsl312l5scc7dnnoxrto 192.168.65.3:2377
This node joined a swarm as a worker.
/# exit
PS C:\Users\sreev> docker exec -it workerc sh
/# docker swarm join --token SWMTKN-1-5bgn26n2382oon5thp3dw6kljnx1nyhp2xaaupkyznftckj8-bs9z8hsl312l5scc7dnnoxrto 192.168.65.3:2377
This node joined a swarm as a worker.
/# exit
PS C:\Users\sreev> docker node ls


| ID           | HOSTNAME       | STATUS | AVAILABILITY | MANAGER STATUS | ENGINE VERSION |
|--------------|----------------|--------|--------------|----------------|----------------|
| b8zwj9xroqat | docker-desktop | Ready  | Active       | Leader         | 29.4.1         |
| 1uohap99j3u5 | docker-desktop | Ready  | Active       |                | 29.4.1         |
| 6ff618d17e5a | docker-desktop | Ready  | Active       |                | 29.4.1         |
| c5ca2cc864d6 | docker-desktop | Ready  | Active       |                | 29.2.1         |


RAM 4.45 GB CPU 0.33% Disk: 69.10 GB used (limit 1006.85 GB)
```

```
Terminal
/# exit
PS C:\Users\sreev> docker node ls


| ID                          | HOSTNAME       | STATUS | AVAILABILITY | MANAGER STATUS | ENGINE VERSION |
|-----------------------------|----------------|--------|--------------|----------------|----------------|
| cew4tjpkjy6y6elacd2te60ux   | 6ff618d17e5a   | Ready  | Active       |                | 29.4.1         |
| 23w5lh97msvmqtcx4q6gqdt     | c5ca2cc864d6   | Ready  | Active       |                | 29.4.1         |
| vpo87xg232tywt2gg322bt9e4 * | docker-desktop | Ready  | Active       | Leader         | 29.2.1         |


PS C:\Users\sreev> docker service ps web


| ID           | NAME  | IMAGE        | NODE           | DESIRED STATE | CURRENT STATE          | ERROR | PORTS |
|--------------|-------|--------------|----------------|---------------|------------------------|-------|-------|
| 1uohap99j3u5 | web.1 | nginx:latest | docker-desktop | Running       | Running 20 minutes ago |       |       |
| ukcnrhufogxr | web.2 | nginx:latest | docker-desktop | Running       | Running 20 minutes ago |       |       |
| pg50q7wq4wf  | web.3 | nginx:latest | docker-desktop | Running       | Running 20 minutes ago |       |       |


PS C:\Users\sreev> docker service update --force web
```

```
Terminal
PS C:\Users\sreev> docker service update --force web
web
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service web converged
```

```
Terminal
ID HOSTNAME STATUS AVAILABILITY MANAGER STATUS ENGINE VERSION
cew4tJpkyJ6yelaclk2te60ux 6ff610d17e5a Ready Active 29.4.1
23vw5Lhf97nsvmqtcx4g6qgdt c5ca2cc864d6 Ready Active 29.4.1
vpo87xg232tywt2gg322bt9e4 * docker-desktop Ready Active Leader 29.2.1

PS C:\Users\sreev> docker service ps web
ID NAME IMAGE NODE DESIRED STATE CURRENT STATE ERROR PORTS
t729k9x1lq6c web.1 nginx:latest docker-desktop Running Running 41 seconds ago
1uohap99j3u5 \ web.1 nginx:latest docker-desktop Shutdown Shutdown 42 seconds ago
83lp7nz18r4e web.2 nginx:latest c5ca2cc864d6 Running Running 46 seconds ago
ukcnrhufogxr \ web.2 nginx:latest docker-desktop Shutdown Shutdown 57 seconds ago
sktghhi0h147 web.3 nginx:latest 6ff610d17e5a Running Running about a minute ago
pg59q7vwq4wf \ web.3 nginx:latest docker-desktop Shutdown Shutdown about a minute ago

RAM 4.44 GB CPU 12.33% Disk 69.10 GB used (limit 1006.85 GB) >_ Terminal Update available
```

```
Terminal
PS C:\Users\sreev> docker service scale web=6
web scaled to 6
overall progress: 6 out of 6 tasks
1/6: running [=====>]
2/6: running [=====>]
3/6: running [=====>]
4/6: running [=====>]
5/6: running [=====>]
6/6: running [=====>]
verify: Service web converged
PS C:\Users\sreev> docker service ps web
ID NAME IMAGE NODE DESIRED STATE CURRENT STATE ERROR PORTS
t729k9x1lq6c web.1 nginx:latest docker-desktop Running Running 3 minutes ago
1uohap99j3u5 \ web.1 nginx:latest docker-desktop Shutdown Shutdown 3 minutes ago
83lp7nz18r4e web.2 nginx:latest c5ca2cc864d6 Running Running 3 minutes ago
ukcnrhufogxr \ web.2 nginx:latest docker-desktop Shutdown Shutdown 4 minutes ago
sktghhi0h147 web.3 nginx:latest 6ff610d17e5a Running Running 4 minutes ago
pg59q7vwq4wf \ web.3 nginx:latest docker-desktop Shutdown Shutdown 4 minutes ago
jpxnj148esat web.4 nginx:latest 6ff610d17e5a Running Running 13 seconds ago
bd9514k2ffgk web.5 nginx:latest c5ca2cc864d6 Running Running 13 seconds ago
w43jgsjypb8n web.6 nginx:latest docker-desktop Running Running 13 seconds ago

RAM 4.45 GB CPU 0.58% Disk 69.10 GB used (limit 1006.85 GB) >_ Terminal Update available
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```
Terminal
PS C:\Users\sreev> docker service ps web
ID NAME IMAGE NODE DESIRED STATE CURRENT STATE ERROR PORTS
t729k9x1lq6c web.1 nginx:latest docker-desktop Running Running 3 minutes ago
1uohap99j3u5 \ web.1 nginx:latest docker-desktop Shutdown Shutdown 3 minutes ago
83lp7nz18r4e web.2 nginx:latest c5ca2cc864d6 Running Running 3 minutes ago
ukcnrhufogxr \ web.2 nginx:latest docker-desktop Shutdown Shutdown 4 minutes ago
sktghhi0h147 web.3 nginx:latest 6ff610d17e5a Running Running 4 minutes ago
pg59q7vwq4wf \ web.3 nginx:latest docker-desktop Shutdown Shutdown 4 minutes ago
jpxnj148esat web.4 nginx:latest 6ff610d17e5a Running Running 13 seconds ago
bd9514k2ffgk web.5 nginx:latest c5ca2cc864d6 Running Running 13 seconds ago
w43jgsjypb8n web.6 nginx:latest docker-desktop Running Running 13 seconds ago
```

☐	☐	web.1.1uohap99j3u5boa0m2wf71	83157b19f169	nginx:latest	0%	31 minutes ago				
☐	☒	workerb	6ff610d17e5a	docker:dind	0.86%	13 minutes ago				
☐	☒	workerc	c5ca2cc864d6	docker:dind	0.78%	13 minutes ago				
☐	☒	web.1.t729k9x1lq6cs4q1wtyh03o	0c684bd13a20	nginx:latest	0%	10 minutes ago				
☐	☒	web.6.w43jgsjypb8mc5xruebah0	6176c6c9851a	nginx:latest	0%	6 minutes ago				
☐	☒	docker-comp3	-	-	0%	7 days ago				

Showing 58 items

☐	☐	web.1.1uohap99j3u5boa0m2wf71	83157b19f169	nginx:latest	0%	32 minutes ago				
☐	☒	workerb	6ff610d17e5a	docker:dind	1.35%	14 minutes ago				
☐	☒	workerc	c5ca2cc864d6	docker:dind	1.01%	14 minutes ago				
☐	☒	web.1.t729k9x1lq6cs4q1wtyh03o	0c684bd13a20	nginx:latest	0%	10 minutes ago				
☐	☒	web.6.w43jgsjypb8mc5xruebah0	6176c6c9851a	nginx:latest	0%	7 minutes ago				
☐	☒	docker-comp3	-	-	0%	7 days ago				

Showing 58 items

Terminal

PS C:\Users\sreev> docker exec -it workerb sh
/ # docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
595ecabbf09f nginx:latest "/docker-entrypoint..." 6 minutes ago Up 6 minutes 88/tcp web.4.jpxnj148esatx7pvlwk2gp68
13658a2c8532 nginx:latest "/docker-entrypoint..." 10 minutes ago Up 10 minutes 88/tcp web.3.sktghhi0h1476b6nw6pluntel
/ # exit
PS C:\Users\sreev> docker exec -it workerc sh
/ # docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
a5fdb53a2863 nginx:latest "/docker-entrypoint..." 6 minutes ago Up 6 minutes 88/tcp web.5.bd9514k2ffgkzndx9vpc1zkr2
79fbd814d9d nginx:latest "/docker-entrypoint..." 10 minutes ago Up 10 minutes 88/tcp web.2.83lp7nz18r4eojyaltev991hc
/ #

RAM 4.45 GB CPU 0.17% Disk 69.10 GB used (limit 1006.85 GB) >_ Terminal Update available

6 replicas. 3 nodes. 1 system that never breaks